Project planning report

Saved project inputs and selected calculation results

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Project

This report presents saved project inputs and the original results of the selected calculations. Country-specific requirements, electrical design and installation checks remain open. It does not approve the complete installation.

Project	Berlin · Beispielprojekt · PV und Speicher
Project reference	c7795bc2c1e14e389622015c809ace45
Input revision	22

Site and project inputs

Location	Berlin · Beispielstandort
Country	DE
Time zone	Europe/Berlin
Project note	Interaktives Beispiel mit echten JRC-Wetterdaten und nachgewiesenen Trina-Moduldaten; Dachgeometrie und Verbrauch sind Beispielannahmen.

PV wiring plan

The wiring plan records the assigned roof modules. Cold open-circuit voltage uses the stated ambient temperature and the module's voltage coefficient. STC operating points do not establish hot or cold MPP performance, design currents or wiring protection.

Lowest design ambient temperature	-10 °C
Temperature source and location	Example condition: -10 °C for this wiring example; the local design extreme has not been verified.

String	MPPT	Input	Modules	Vmpp at STC	Cold Voc
1	1	1	12	544.8 V	699.83 V
2	1	2	12	544.8 V	699.83 V

Modeled electrical operating points

These estimates use a fitted single-diode model at the stated uniform module conditions. They do not include shading or establish manufacturer performance, design extremes, protection ratings or country approval.

Illustrative uniform operating cases for this saved example: effective irradiance 1000 W/m²; hot cell 60 °C, cold cell 20 °C. Mounting, ventilation, wind and local design extremes have not been established. These are entered scenario conditions, not site measurements or manufacturer release.

Effective irradiance	1000 W/m²
Hot operating cell temperature	60 °C
Cold operating cell temperature	20 °C

String	Case	Vmpp	Impp	Isc
1	Hot operating case	487.173 V	10.2021 A	10.9611 A
1	Cold operating case	553.062 V	10.1291 A	10.7884 A
2	Hot operating case	487.173 V	10.2021 A	10.9611 A
2	Cold operating case	553.062 V	10.1291 A	10.7884 A

Model, inputs and datasheet comparisons

Batzelis initialization and De Soto fitting use the saved STC points and temperature coefficients. The fitted compound parameters do not identify the physical cell-series count. Numerical fit precision is separate from physical accuracy.

pvlib 0.16.1

Source document · Page 2	https://de-linker.com/wp-content/uploads/2026/09/240831_Datasheet_Vertex-S_NEG9R.28_EN_2024_C_web.pdf https://de-linker.com/wp-content/uploads/2026/09/240831_Datasheet_Vertex-S_NEG9R.28_EN_2024_C_web.pdf
Fitting method	Batzelis https://pvlib-python.readthedocs.io/en/v0.16.1/reference/generated/pvlib.ivtools.sdm.fit_desoto_batzelis.html
Numerical refinement	De Soto https://pvlib-python.readthedocs.io/en/v0.16.1/reference/generated/pvlib.ivtools.sdm.fit_desoto.html
Temperature translation	De Soto https://pvlib-python.readthedocs.io/en/v0.16.1/reference/generated/pvlib.pvsystem.calcparams_desoto.html
Model reference	cdba01f814f63187c5b485df22a93a2c7c7409585535f21ed3fddac086a72dfa

Quantity	Value
□Isc	0.04 %/°C
□Voc	-0.24 %/°C
Cell technology	N-type i-TOPCon monocrystalline
Impp (STC)	10.14 A
Isc (STC)	10.81 A
Vmpp (STC)	45.4 V
Voc (STC)	53.8 V

Fitted model parameters

a,ref	1.84665 V
IL,ref	10.8242 A
I0,ref	2.34402e-12 A
Rs	0.244312 Ω
Rsh,ref	185.625 Ω
□Isc	0.004324 A/K
Silicon bandgap assumption	1.121 eV
Bandgap temperature assumption	-0.0002677 1/K
Relative numerical residual	3.8848e-11
Numerical fit tolerance	1.49012e-08

Datasheet comparisons

The power coefficient and NOCT values were not fitting targets. These comparisons show differences; they do not certify extrapolation accuracy.

Declared power temperature coefficient	-0.29 %/°C
Model power temperature coefficient	-0.282132 %/°C
NOCT reference irradiance	800 W/m²
NOCT cell-temperature interval	41 °C – 45 °C

Declared NOCT cell temperature	43 °C
Declared NOCT power	352 W
Model power at nominal NOCT	349.996 W
Model powers at the NOCT temperature limits	352.127 W / 347.861 W

Equipment checks

Country requirements are not assessed by the selected calculations.

Assessment status	Further checks required

Converter output

Assessment status	Checked requirements met
Result	10 kW
Source document · Page 2	10000 W https://de-linker.com/wp-content/uploads/2026/09/H2-10-30K-T2T3-datasheet-EN-BEU.pdf

Grid phases

Assessment status	Checked requirements met
Result	3
Requirement	3

PV input power

Assessment status	Checked requirements met
Result	20 kWp
Requirement	11.04 kWp
Source document · Page 2	20000 Wp https://de-linker.com/wp-content/uploads/2026/09/H2-10-30K-T2T3-datasheet-EN-BEU.pdf

Roof module assignment

Assessment status	Checked requirements met
Result	24
Requirement	24

String 1 · MPPT assignment

Assessment status	Checked requirements met
Result	1
Requirement	3
Source document · Page 2	3 https://de-linker.com/wp-content/uploads/2026/09/H2-10-30K-T2T3-datasheet-EN-BEU.pdf

String 1 · Inverter input assignment

The input number identifies the proposed connection. A cited manufacturer connector map is still required.

Assessment status	Further checks required
Result	1

String 1 · Cold open-circuit voltage

Voc = modules per string × module Voc at STC × [1 + □Voc × (lowest design ambient temperature − 25 °C)]. □Voc is the relative Voc temperature coefficient in 1/°C; divide a value in %/°C by 100. The design temperature and its source are required.

Assessment status	Checked requirements met
Result	699.83 V
Requirement	1,000 V
Source document · Page 2	53.8 V https://de-linker.com/wp-content/uploads/2026/09/240831_Datasheet_Vertex-S_NEG9R.28_EN_2024_C_web.pdf
Source document · Page 2	1000 V https://de-linker.com/wp-content/uploads/2026/09/H2-10-30K-T2T3-datasheet-EN-BEU.pdf

String 1 · Module system voltage (IEC rating)

Assessment status	Checked requirements met
Result	699.83 V
Requirement	1,500 V
Source document · Page 2	53.8 V https://de-linker.com/wp-content/uploads/2026/09/240831_Datasheet_Vertex-S_NEG9R.28_EN_2024_C_web.pdf

String 1 · MPP voltage at STC

This comparison uses STC values: 1,000 W/m² irradiance, 25 °C cell temperature and the AM1.5 reference spectrum.

Assessment status	Checked requirements met
Result	544.8 V
Requirement	180 – 900 V
Source document · Page 2	45.4 V https://de-linker.com/wp-content/uploads/2026/09/240831_Datasheet_Vertex-S_NEG9R.28_EN_2024_C_web.pdf
Source document · Page 2	180~900 V https://de-linker.com/wp-content/uploads/2026/09/H2-10-30K-T2T3-datasheet-EN-BEU.pdf

String 2 · MPPT assignment

Assessment status	Checked requirements met
Result	1
Requirement	3
Source document · Page 2	3 https://de-linker.com/wp-content/uploads/2026/09/H2-10-30K-T2T3-datasheet-EN-BEU.pdf

String 2 · Inverter input assignment

The input number identifies the proposed connection. A cited manufacturer connector map is still required.

Assessment status	Further checks required
Result	2

String 2 · Cold open-circuit voltage

Voc = modules per string × module Voc at STC × [1 + □Voc × (lowest design ambient temperature − 25 °C)]. □Voc is the relative Voc temperature coefficient in 1/°C; divide a value in %/°C by 100. The design temperature and its source are required.

Assessment status	Checked requirements met
Result	699.83 V
Requirement	1,000 V
Source document · Page 2	53.8 V https://de-linker.com/wp-content/uploads/2026/09/240831_Datasheet_Vertex-S_NEG9R.28_EN_2024_C_web.pdf
Source document · Page 2	1000 V https://de-linker.com/wp-content/uploads/2026/09/H2-10-30K-T2T3-datasheet-EN-BEU.pdf

String 2 · Module system voltage (IEC rating)

Assessment status	Checked requirements met
Result	699.83 V
Requirement	1,500 V
Source document · Page 2	53.8 V https://de-linker.com/wp-content/uploads/2026/09/240831_Datasheet_Vertex-S_NEG9R.28_EN_2024_C_web.pdf

String 2 · MPP voltage at STC

This comparison uses STC values: 1,000 W/m² irradiance, 25 °C cell temperature and the AM1.5 reference spectrum.

Assessment status	Checked requirements met
Result	544.8 V
Requirement	180 – 900 V
Source document · Page 2	45.4 V https://de-linker.com/wp-content/uploads/2026/09/240831_Datasheet_Vertex-S_NEG9R.28_EN_2024_C_web.pdf
Source document · Page 2	180~900 V https://de-linker.com/wp-content/uploads/2026/09/H2-10-30K-T2T3-datasheet-EN-BEU.pdf

Wired PV nameplate power

Assessment status	Checked requirements met
Result	11.04 kWp
Requirement	11.04 kWp

MPPT 1 · Strings per MPPT

Assessment status	Checked requirements met
Result	2
Requirement	2
Source document · Page 2	2 https://de-linker.com/wp-content/uploads/2026/09/H2-10-30K-T2T3-datasheet-EN-BEU.pdf

MPPT 1 · MPPT operating current at STC

Parallel-string currents are added at STC. Design irradiance, temperature and applicable rating factors require separate checks.

Assessment status	Checked requirements met
Result	20.28 A
Requirement	40 A
Source document · Page 2	10.14 A https://de-linker.com/wp-content/uploads/2026/09/240831_Datasheet_Vertex-S_NEG9R.28_EN_2024_C_web.pdf
Source document · Page 2	40 A https://de-linker.com/wp-content/uploads/2026/09/H2-10-30K-T2T3-datasheet-EN-BEU.pdf

MPPT 1 · MPPT short-circuit current at STC

Parallel-string currents are added at STC. Design irradiance, temperature and applicable rating factors require separate checks.

Assessment status	Checked requirements met
Result	21.62 A
Requirement	50 A
Source document · Page 2	10.81 A https://de-linker.com/wp-content/uploads/2026/09/240831_Datasheet_Vertex-S_NEG9R.28_EN_2024_C_web.pdf
Source document · Page 2	50 A https://de-linker.com/wp-content/uploads/2026/09/H2-10-30K-T2T3-datasheet-EN-BEU.pdf

MPPT 1 · Parallel string lengths and orientations

Assessment status	Checked requirements met

Single-diode model fit

The fitted parameters reproduce the supplied STC points within the stated numerical tolerance. This is a numerical check, not a manufacturer performance validation.

Assessment status	Checked requirements met
Source document · Page 2	45.4 V https://de-linker.com/wp-content/uploads/2026/09/240831_Datasheet_Vertex-S_NEG9R.28_EN_2024_C_web.pdf

Model applicability to this module

The fit does not establish model accuracy for this exact module over its operating range. Validation against manufacturer performance data remains required.

Assessment status	Further checks required

Operating-condition domain

The calculation applies the stated effective irradiance uniformly to the modules at each declared cell temperature.

Assessment status	Checked requirements met
Requirement	-40 – 85 °C
Source document · Page 2	-40 to +85 °C https://de-linker.com/wp-content/uploads/2026/09/240831_Datasheet_Vertex-S_NEG9R.28_EN_2024_C_web.pdf

String 1 · Modeled MPP voltage: hot case

This compares the modeled MPP voltage at the declared case with the documented MPPT window. Other irradiance, temperature and shading conditions require their own assessment.

Assessment status	Checked requirements met
Result	487.173 V
Requirement	180 – 900 V
Source document · Page 2	45.4 V https://de-linker.com/wp-content/uploads/2026/09/240831_Datasheet_Vertex-S_NEG9R.28_EN_2024_C_web.pdf
Source document · Page 2	180~900 V https://de-linker.com/wp-content/uploads/2026/09/H2-10-30K-T2T3-datasheet-EN-BEU.pdf

String 1 · Modeled MPP voltage: cold case

This compares the modeled MPP voltage at the declared case with the documented MPPT window. Other irradiance, temperature and shading conditions require their own assessment.

Assessment status	Checked requirements met
Result	553.062 V
Requirement	180 – 900 V
Source document · Page 2	45.4 V https://de-linker.com/wp-content/uploads/2026/09/240831_Datasheet_Vertex-S_NEG9R.28_EN_2024_C_web.pdf
Source document · Page 2	180~900 V https://de-linker.com/wp-content/uploads/2026/09/H2-10-30K-T2T3-datasheet-EN-BEU.pdf

String 2 · Modeled MPP voltage: hot case

This compares the modeled MPP voltage at the declared case with the documented MPPT window. Other irradiance, temperature and shading conditions require their own assessment.

Assessment status	Checked requirements met
Result	487.173 V
Requirement	180 – 900 V
Source document · Page 2	45.4 V https://de-linker.com/wp-content/uploads/2026/09/240831_Datasheet_Vertex-S_NEG9R.28_EN_2024_C_web.pdf
Source document · Page 2	180~900 V https://de-linker.com/wp-content/uploads/2026/09/H2-10-30K-T2T3-datasheet-EN-BEU.pdf

String 2 · Modeled MPP voltage: cold case

This compares the modeled MPP voltage at the declared case with the documented MPPT window. Other irradiance, temperature and shading conditions require their own assessment.

Assessment status	Checked requirements met
Result	553.062 V
Requirement	180 – 900 V
Source document · Page 2	45.4 V https://de-linker.com/wp-content/uploads/2026/09/240831_Datasheet_Vertex-S_NEG9R.28_EN_2024_C_web.pdf

Source document · Page 2	180~900 V https://de-linker.com/wp-content/uploads/2026/09/H2-10-30K-T2T3-datasheet-EN-BEU.pdf

MPPT 1 · Modeled MPPT current: hot case

Parallel currents are modeled at the declared uniform case. These points do not replace design-current or protection-rating checks.

Assessment status	Checked requirements met
Result	20.404 A
Requirement	40 A
Source document · Page 2	45.4 V https://de-linker.com/wp-content/uploads/2026/09/240831_Datasheet_Vertex-S_NEG9R.28_EN_2024_C_web.pdf
Source document · Page 2	40 A https://de-linker.com/wp-content/uploads/2026/09/H2-10-30K-T2T3-datasheet-EN-BEU.pdf

MPPT 1 · Modeled short-circuit current: hot case

Parallel currents are modeled at the declared uniform case. These points do not replace design-current or protection-rating checks.

Assessment status	Checked requirements met
Result	21.922 A
Requirement	50 A
Source document · Page 2	45.4 V https://de-linker.com/wp-content/uploads/2026/09/240831_Datasheet_Vertex-S_NEG9R.28_EN_2024_C_web.pdf
Source document · Page 2	50 A https://de-linker.com/wp-content/uploads/2026/09/H2-10-30K-T2T3-datasheet-EN-BEU.pdf

MPPT 1 · Modeled MPPT current: cold case

Parallel currents are modeled at the declared uniform case. These points do not replace design-current or protection-rating checks.

Assessment status	Checked requirements met
Result	20.258 A
Requirement	40 A
Source document · Page 2	45.4 V https://de-linker.com/wp-content/uploads/2026/09/240831_Datasheet_Vertex-S_NEG9R.28_EN_2024_C_web.pdf
Source document · Page 2	40 A https://de-linker.com/wp-content/uploads/2026/09/H2-10-30K-T2T3-datasheet-EN-BEU.pdf

MPPT 1 · Modeled short-circuit current: cold case

Parallel currents are modeled at the declared uniform case. These points do not replace design-current or protection-rating checks.

Assessment status	Checked requirements met
Result	21.577 A
Requirement	50 A

Source document · Page 2	45.4 V https://de-linker.com/wp-content/uploads/2026/09/240831_Datasheet_Vertex-S_NEG9R.28_EN_2024_C_web.pdf
Source document · Page 2	50 A https://de-linker.com/wp-content/uploads/2026/09/H2-10-30K-T2T3-datasheet-EN-BEU.pdf

Design current

Design currents require documented irradiance, temperature and applicable rating rules.

Assessment status	Further checks required

Wiring and protection

Cable routes, conductor ratings, connector compatibility, fuses and isolation still require project-specific design.

Assessment status	Further checks required

Required accessory

Assessment status	Checked requirements met
Result	SAJ BC2-HV1

Manufacturer pairing

Assessment status	Further checks required

Module count

Assessment status	Checked requirements met
Result	4
Requirement	2 – 5

Parallel arrangement

Assessment status	Checked requirements met
Result	1

Battery voltage

Assessment status	Checked requirements met
Source document · Page 2	180~800 V https://de-linker.com/wp-content/uploads/2026/09/H2-10-30K-T2T3-datasheet-EN-BEU.pdf
Source document · Page 2	89.6-115.2 / 179.2-230.4 / 268.8-345.6 / 358.4-460.8 / 448-576.0 V https://de-linker.com/wp-content/uploads/2026/09/B2-5-25K-HV1HV5-datasheet-EN-BEU.pdf

Sales unit

Assessment status	Checked requirements met
Source document · Page 2	battery_module https://de-linker.com/wp-content/uploads/2026/09/B2-5-25K-HV1HV5-datasheet-EN-BEU.pdf

Usable energy

4.5 × 4 = 18 kWh

Assessment status	Checked requirements met
Result	18 kWh
Requirement	10 kWh
Source document · Page 2	4.5 kWh https://de-linker.com/wp-content/uploads/2026/09/B2-5-25K-HV1HV5-datasheet-EN-BEU.pdf

Continuous AC discharge power

Assessment status	Further checks required
Requirement	5 kW

Battery discharge path

Assessment status	Further checks required

Components and manufacturer conditions

The H2 T3 ships with a smart meter, three current transformers and a 10 m meter communication cable (H2 manual, section 2.2 Unpacking, PDF p. 36 English / p. 7 German).

SAJ's B2 datasheet names B2-5.0 to B2-25.0 as one to five BU2-5.0 modules with one BC2-HV1 control module, and the B2 manual (V1.2, PDF p5) says the B2 can only be used as such a set: the control module is an article of its own.

SAJ's Configuration Instructions V1.0 (Section 2) give the H2 series system as H2 + BC2 + BU2. SAJ's B2 datasheet names B2-5.0 to B2-25.0 as one to five BU2-5.0 modules with one BC2-HV1 control module, and SAJ's B2 manual (V1.2, PDF p5) says the B2 can only be used as a set with SAJ's H2 high voltage series storage inverter. The listing does not cover the one-module B2-5.0, whose operating voltage lies below the H2 T3's battery voltage range.

One BU2-5.0 module alone operates at 89.6 to 115.2 V (SAJ's B2 datasheet gives this for the one-module B2-5.0), below the H2 T3's battery voltage range of 180 to 800 V; two modules operate at 179.2 to 230.4 V.

Source document · Page 36	https://de-linker.com/wp-content/uploads/2026/10/SAJ_H2_10-30K_User-Manual_DE-EN_EU.pdf https://de-linker.com/wp-content/uploads/2026/10/SAJ_H2_10-30K_User-Manual_DE-EN_EU.pdf
Source document	https://de-linker.com/wp-content/uploads/2026/09/H2_HS2_HS3_AS2_AS3_Configuration_Instructions_V1.0_en.pdf https://de-linker.com/wp-content/uploads/2026/09/H2_HS2_HS3_AS2_AS3_Configuration_Instructions_V1.0_en.pdf
Source document	https://de-linker.com/wp-content/uploads/2026/10/B2-5.0-HV1-HV5-user-manual-EN-EU.pdf https://de-linker.com/wp-content/uploads/2026/10/B2-5.0-HV1-HV5-user-manual-EN-EU.pdf
Source document · Page 2	https://de-linker.com/wp-content/uploads/2026/09/B2-5-25K-HV1HV5-datasheet-EN-BEU.pdf https://de-linker.com/wp-content/uploads/2026/09/B2-5-25K-HV1HV5-datasheet-EN-BEU.pdf
Source document · Page 37	https://de-linker.com/wp-content/uploads/2026/10/SAJ_H2_10-30K_User-Manual_DE-EN_EU.pdf https://de-linker.com/wp-content/uploads/2026/10/SAJ_H2_10-30K_User-Manual_DE-EN_EU.pdf

Provisional bill of materials

This provisional list comes from the selected saved equipment result. Quantities are shown only where established. It is not a complete installation bill of materials. Further components and open requirements need project-specific assessment.

Quantity	Product	Reference
1	SAJ H2-10K-T3	DLK-SAJ-H210KT3
1	SAJ BC2-HV1	DLK-SAJ-BC2HV1
4	SAJ BU2-5.0-HV1	DLK-SAJ-BU250HV1

PV modules used in the calculation

Module counts come from the saved layout of the selected generation result. The listed modules are design references; electrical and mounting suitability require project-specific assessment.

This list includes only the PV conversion group used by the selected DC calculation.

Input revision	9
PV nameplate power	11.04 kWp
Source document · Page 2	Trina Solar TSM-460NEG9R.28 https://de-linker.com/wp-content/uploads/2026/09/240831_Datasheet_Vertex-S_NEG9R.28_EN_2024_C_web.pdf

Quantity	Model	Module power (STC)
24	Trina Solar TSM-460NEG9R.28	460 W

Energy scenario and selected equipment

These checks compare the saved scenario parameters with the equipment selected for this calculation. They do not establish manufacturer performance.

4.5 × 4 = 18 kWh

Assessment status	Further checks required
Model	SAJ BU2-5.0-HV1
Model	SAJ BC2-HV1
Model	SAJ H2-10K-T3
Module count	4
Parallel arrangement	1
Usable energy · Source document · Page 2	4.5 kWh https://de-linker.com/wp-content/uploads/2026/09/B2-5-25K-HV1HV5-datasheet-EN-BEU.pdf

Check	Assessment status	Result	Requirement
Usable energy	Checked requirements met	18 kWh	18 kWh
Continuous AC discharge power	Further checks required	—	10 kW
Scenario operating mode	Checked requirements met	—	—
Storage losses and operating conditions	Further checks required	—	—
Converter ports and system topology	Further checks required	—	—

Energy scenario

PV generation and curtailment are calculated at the DC bus; load supply and grid flows are calculated on the AC side. Storage results use the saved loss assumptions.

Battery temperature is a fixed scenario condition used to check the documented charging and discharging temperature ranges. It is not inferred from outdoor weather. This model uses fixed loss assumptions and does not simulate temperature changes.

Simulated load demand	1.103 kWh
PV generation (DC)	9.634 kWh

Grid import	0 kWh
PV export	0 kWh
PV curtailment (DC)	0 kWh
Unserved simulated load	0 kWh
Solar share of gross demand	100 %
PV used or stored on site	100 %
Calculated intervals	2
Battery temperature for this scenario	20 °C

Saved calculation references

Data source	Beispielannahme · kein externer AC-Hilfsverbrauch
Data source	Declared reference assumptions: DC/DC 98% each way, storage 99% each way, forward inverter 96%, fixed battery temperature 20 °C, zero internal and DC auxiliary demand and standing loss. No reverse charging. These are scenario assumptions, not manufacturer performance data.
Data source	Beispielverbrauch · 6.000 kWh pro Jahr
Data source	JRC PVGIS typical meteorological year
Calculation reference	6d5c7169efc24bb297691f23964daf4c · Input revision 22
Calculation reference	4a89969354884bb79292f9ca000574c4 · Input revision 20